

```

print("NAME: KAVYA S USN:1BM24CS131")
def vacuum_world():
    # 0 indicates CLEAN, 1 indicates DIRTY
    goal_state = {'A': '0', 'B': '0'}
    cost = 0

    print("--- Vacuum Cleaner Environment Setup ---")
    # 1. Get initial vacuum location
    location_input = input("Enter initial location of Vacuum (A/B): ").upper().strip()

    # 2. Get the status of the rooms
    status_input = input(f"Enter status of Room {location_input} (0 for CLEAN, 1 for DIRTY): ").strip()

    other_room = 'B' if location_input == 'A' else 'A'
    status_input_complement = input(f"Enter status of Room {other_room} (0 for CLEAN, 1 for DIRTY): ").strip()

    # Create current environment dictionary
    environment = {location_input: status_input, other_room: status_input_complement}

    print(f"\nInitial Environment State: {environment}")
    print("Starting simulation...")

    # Process Room A
    if location_input == 'A':
        # Condition for Room A
        if environment['A'] == '1':
            print("Room A is DIRTY. Sucking the dirt...")
            environment['A'] = '0'
            cost += 1 # Cost for cleaning
            print(f"Room A cleaned. Current Cost: {cost}")
        else:
            print("Room A is already CLEAN.")

        # Move to Room B
        print("Moving to Room B...")
        cost += 1 # Cost for moving

        # Condition for Room B
        if environment['B'] == '1':
            print("Room B is DIRTY. Sucking the dirt...")
            environment['B'] = '0'
            cost += 1 # Cost for cleaning
            print(f"Room B cleaned. Current Cost: {cost}")
        else:
            print("Room B is already CLEAN.")

    # Process Room B
    else:
        # Condition for Room B
        if environment['B'] == '1':
            print("Room B is DIRTY. Sucking the dirt...")
            environment['B'] = '0'
            cost += 1 # Cost for cleaning
            print(f"Room B cleaned. Current Cost: {cost}")
        else:
            print("Room B is already CLEAN.")

        # Move to Room A
        print("Moving to Room A...")
        cost += 1 # Cost for moving

        # Condition for Room A
        if environment['A'] == '1':
            print("Room A is DIRTY. Sucking the dirt...")
            environment['A'] = '0'
            cost += 1 # Cost for cleaning
            print(f"Room A cleaned. Current Cost: {cost}")
        else:
            print("Room A is already CLEAN.")

    # Final Summary
    print("\n--- Simulation Results ---")
    print(f"Final Environment State: {environment}")
    print(f"Goal State Met: {environment == goal_state}")
    print(f"Total Path Cost: {cost}")

# Run the program
if __name__ == "__main__":
    vacuum_world()

```

```
NAME: KAVYA S USN:1BM24CS131
--- Vacuum Cleaner Environment Setup ---
Enter initial location of Vacuum (A/B): B
Enter status of Room B (0 for CLEAN, 1 for DIRTY): 1
Enter status of Room A (0 for CLEAN, 1 for DIRTY): 1

Initial Environment State: {'B': '1', 'A': '1'}
Starting simulation...
Room B is DIRTY. Sucking the dirt...
Room B cleaned. Current Cost: 1
Moving to Room A...
Room A is DIRTY. Sucking the dirt...
Room A cleaned. Current Cost: 3

--- Simulation Results ---
Final Environment State: {'B': '0', 'A': '0'}
Goal State Met: True
Total Path Cost: 3
```